

Sebastian Flores

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EDUCATION

Georgia Institute of Technology

Master of Science (MS) in Mechanical Engineering

Atlanta, GA

Expected May 2028

Bachelor of Science (BS) in Mechanical Engineering

Expected May 2027

GPA: 4.0

- Concentration in Automation & Robotic Systems
- Relevant Coursework: Controls, System Dynamics, Machine Design, Design for Manufacture, Heat Transfer

TECHNICAL SKILLS

Software: SolidWorks, MATLAB, Java, C++, Python, Yaskawa MotoSim, FANUC RoboGuide, Rockwell Studio 5000

Robotics: FANUC, Yaskawa, Denso, PLC Programming, Apera 3D Vision, Kinematics Simulation

Hardware & Fabrication: 3D Printing (FDM/SLA), DFM/DFA, Soldering, Laser Cutting, Shop Tools, Waterjet

EXPERIENCE

Advanced Robotics for Manufacturing (ARM) Institute

Pittsburgh, PA

Mechanical Engineering Intern

May – Aug 2026

- Integrated a FANUC cobot with Apera 3D vision to bin-pick cast steel parts for drill press tending
- Implemented vision-based pose-checking algorithms, increasing part placement accuracy by 60%
- Developed custom part fixtures with embedded magnets to compensate for inconsistent part pick positions
- Optimized program logic for concurrent part-processing and bin-scanning, reducing cycle time by 40%
- Simulated a pipe-spool welding cell in Yaskawa MotoSim, developing parametric welding routines to accommodate pipes up to 3m; validated cell layout and concept feasibility for a major client presentation
- Designed and prototyped modular end-of-arm tooling (EOAT) to package bullet cartridges; applied DFM/DFA principles to transition 3D-printed prototypes into machined assemblies with quick-release components

Flowers Invention Studio

Atlanta, GA

Prototyping Instructor

Dec 2025 – Present

- Supervise, train, and mentor students and researchers in the nation's largest student-run makerspace
- Provide 1-on-1 technical support for FDM/SLA 3D printers, electronics tools, and machine shop equipment
- Guide users through standard operating procedures (SOPs) to mitigate hazards and promote safety

GT Marine Robotics Group

Atlanta, GA

Mechanical Engineering Subteam

Jan 2025 – Present

- Constructed a small, autonomous, robotic boat; integrated sensors and actuators using 3D-printed brackets
- Redesigned internal layout of RoboSub competition vehicle; collaborated with electrical subteam to design and fabricate a quick-disconnect wiring plate, improving access to internals and reducing maintenance time by 50%
- Designing and manufacturing a custom keel for the SailBot competition vehicle utilizing SLA 3D printing, filling the internal cavity with a lead-shot-and-resin composite to lower the center of mass and improve stability

Cosmic Constructors – Space Construction Robot

Sep – Dec 2024

- Modeled 9 mechanical parts for a hopper-and-feeder subassembly to integrate with teammates' constructions
- Drafted scale drawings for parts and rendered photorealistic mock-up images using SolidWorks Visualize

OctoGifts

Jan 2019 – Dec 2021

- Invented a paper candy dispenser and DIY kit that sold over 400 units in 25 states and was awarded 2 patents

LEADERSHIP

Georgia Institute of Technology

Atlanta, GA & Metz, France

Teaching Assistant – Multivariable Calculus, Dynamics, Computational Methods

May 2025 – May 2026

- Led 2 interactive studio sessions per week for 10 students in Multivariable Calculus at GT's Europe campus
- Graded and hosted weekly office hours for 125 students in Dynamics and Computational Methods in Atlanta

Phi Sigma Kappa Fraternity

Atlanta, GA

Secretary

Aug 2024 – Present

- Directed a 5-person team to overhaul chapter bylaws, improving governing document readability
- Implemented centralized record-keeping and financial tracking systems for an 80+ member organization

HONORS, AWARDS, & PATENTS

Hispanic Scholarship Fund Scholar

Jun 2024

US Patent 11,254,486; Apparatus for dispensing objects and method

Iss. Feb 22, 2022

US Patent D923,419; Object dispenser

Iss. Jun 29, 2021